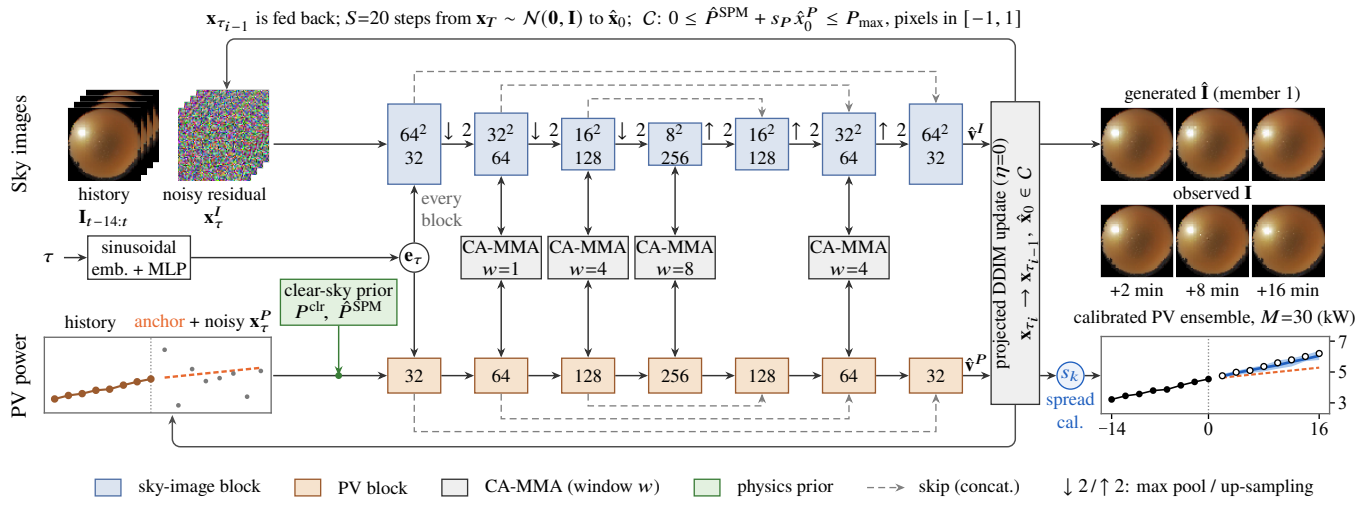

$\mathbf{x}_{\tau_{i-1}}$ is fed back; S=20 steps from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to $\hat{\mathbf{x}}_0$; C: $0 \le \hat{P}^{\mathrm{SPM}} + s_P \hat{x}_0^P \le P_{\max}$, pixels in $[-1, 1]$
Sky images
history $\mathbf{I}_{t-14:t}$
noisy residual $\mathbf{x}_\tau^I$
τ
sinusoidal emb. + MLP
$\mathbf{e}_\tau$
every block
64^2 32
$\downarrow 2$
32^2 64
16^2 128
8^2 256
$\uparrow 2$
CA-MMA w=1
CA-MMA w=4
CA-MMA w=8
CA-MMA w=4
$\hat{\mathbf{v}}^I$
PV power
history
anchor + noisy $\mathbf{x}_\tau^P$
clear-sky prior P^{clr}, $\hat{P}^{\mathrm{SPM}}$
32
64
128
256
$\hat{\mathbf{v}}^P$
projected DDIM update (η=0)
$\mathbf{x}_{\tau_i} \to \mathbf{x}_{\tau_{i-1}}$, $\hat{\mathbf{x}}_0 \in \mathcal{C}$
generated $\hat{\mathbf{I}}$ (member 1)
observed $\mathbf{I}$
+2 min
+8 min
+16 min
calibrated PV ensemble, M=30 (kW)
s_k
spread cal.
−14
0
16
7
5
3
sky-image block
PV block
CA-MMA (window w)
physics prior
skip (concat.)
$\downarrow 2$/$\uparrow 2$: max pool / up-sampling